Daksh Arora

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github.com/dakshcodez

Education

Vellore Institute of Technology, B.Tech in Computer Science and Engineering — Sep 2024 - Jul 2028

- CGPA: 8.87
- Senior Core Committee Member at IEEE-Computer Society VIT

Technical Skills

Languages: Golang, TypeScript, JavaScript, Python, C, C++, Java
Frameworks: Node.js, Express.js, Nest.js, FastAPI, Gin, React.js, TailwindCSS, OpenCV
Tools: Git, Postman, Firebase, Supabase, AWS
Databases: DynamoDB, MySQL, MongoDB, PostgreSQL, MariaDB, Redis, Firebase Firestore

Experience

Software Development Engineer Intern, Independ –remote — June 2026 –Present

- Wrote unit tests for the **files** and **admin** modules covering schedulers, services, and repositories using **Vitest** within a **NestJS** backend, validating **CASL**-based access control, cursor pagination, and **dependency injection** across 1,700+ lines of test code.
- Built an **AI chat and negotiations context** module, enriching **LLM** prompts with authenticated **conversation history** and structured negotiation/bid data (budget, timeline, milestones) via participant-scoped context services, improving response accuracy and relevance.
- Actively building **guardrails** for the AI pipeline – **JSON-schema-constrained structured outputs**, strict **Zod** validation, and automatic retry-on-invalid-output logic – to prevent **hallucinations** and stop the model from returning false or unverified data.

Software Developer Intern, TX Minds –remote — Apr 2026 –Sep 2026

- Built the donation and payment backend for a non-profit platform (**Node.js**, **TypeScript**, **Express 5**, **Drizzle ORM**, **PostgreSQL**), integrating **HDFC** and **ICICI** payment gateways, **2FA**, **Cloudflare Turnstile** bot protection, and a custom **Strapi CMS** OTP-auth plugin for admin login.
- Designed an asynchronous receipt pipeline on **BullMQ** and **Redis** with idempotent, rate-limited jobs and exponential-backoff retries, delivering donation receipts and OTP/password-reset emails through **AWS SES**; extended SES delivery to Strapi admin password resets.
- Led the migration from Render to **AWS** (ap-south-1): containerised the API and CMS on **EC2** behind an **ALB** with **ACM**-managed TLS, **RDS** PostgreSQL over verified TLS, and **S3** for media; wrote the deployment runbook.
- Built a CI/CD pipeline in **GitHub Actions**: type-check, build and dependency-audit gates on every PR, then push-button production deploys to **ECR** and **EC2** via **OIDC** and **SSM** (no stored AWS keys), with health-check gating and automatic rollback.

Projects

Peer-to-Peer File Sharing System — Github Link

- Built a distributed peer-to-peer file sharing system using deterministic file chunking, content-addressable storage, and hash-based integrity verification for fault-tolerant, consistent data distribution across decentralized nodes.
- Developed a custom TCP networking layer with length-prefixed message framing, versioned handshakes, and optimized data transfer workflows for reliable, backward-compatible communication.
- **Technologies:** Golang, TCP Sockets, Distributed Systems

Lore – On-Device AI Knowledge Assistant — Github Link

- Built an on-device AI knowledge assistant during the Qualcomm Snapdragon Multiverse Hackathon, designing a Retrieval-Augmented Generation (RAG) pipeline with document parsing, semantic chunking, vector search, and contextual response generation.
- Optimized inference using Qualcomm AI Hub models on Snapdragon NPUs and engineered a scalable backend for document ingestion, indexing, retrieval, and session management, eliminating cloud dependency while improving latency.
- **Technologies:** Python, FastAPI, RAG, LanceDB, Qualcomm AI Hub, Snapdragon NPU

Achievements

- Winner, **Qualcomm Snapdragon Multiverse Hackathon** –built **Lore**, an on-device AI assistant on Snapdragon NPUs.
- Top 30, **Reverse Coding X**, a reverse coding competition hosted by **IIT Madras** as part of Shaastra.
- Senior Core Committee Member, **IEEE Computer Society VIT** –hackathons, coding contests, and workshops.